Real-Time Cyber Attack Detection via Machine Learning and Deep Learning Approaches

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***Abstract*—The rapid expansion of the cyber world has significantly increased the frequency and complexity of cyber attacks, with network intrusions remaining among the most prevalent and damaging threats. To counter these threats, Intrusion Detection Systems (IDS) have become essential in modern cybersecurity infrastructures. Machine learning (ML) and deep learning (DL) techniques are particularly effective in real-time intrusion detection due to their ability to learn complex patterns from high-dimensional data. This study presents a hybrid IDS framework using ML and DL models, evaluated on the widely adopted and imbalanced UNSW-NB15 dataset, in which only 13 % of the samples correspond to attack traffic. Five ML algorithms were implemented: logistic regression, decision tree, random forest, CatBoost, and XGBoost, along with various Deep Neural Network (DNN) architectures and autoencoders. Due to class imbalance, the evaluation emphasized recall and F1 score over accuracy. Among ML models, XGBoost achieved the best performance, with an F1 score of 98.08%, recall of 97.87%, and precision of 98.30%, while maintaining a low prediction time of 0.60 µs/sample. In DL models, Focal Loss-based DNN effectively addressed minority class detection, attaining an F1 score of 96.80% and recall of 96.63%. These findings highlight the effectiveness of learning-based approaches, especially ensemble models like XGBoost and class-imbalance-aware deep networks like Focal Loss DNN, for real-time intrusion detection.**

***Index Terms*—Intrusion Detection, Machine Learning, Deep Learning, XGBoost, Feature Selection, Anomaly Detection**

I. Introduction

Real-time cyber attack detection has become increasingly critical due to the rapid evolution and sophistication of cyber threats. Machine learning (ML) and deep learning (DL) techniques offer promising solutions to identify malicious behavior in network traffic and application layers, such as phishing detection [1] and malware detection [4]. The exponential growth in network traffic and the increasing complexity of cyber threats have made traditional rule-based intrusion detection systems insufficient. To address this, ML and DL techniques have gained prominence in detecting anomalous behaviors within network environments. The UNSW-NB15 dataset, containing a mixture of modern attack types and normal traffic, offers a suitable benchmark to evaluate such intelligent systems.

Intrusion refers to any unauthorized activity that attempts to compromise the confidentiality, integrity, or availability of a computer system or network [2]. In cybersecurity, IDS are specialized technologies designed to identify and prevent such malicious activities. The growing sophistication of cyber threats, including ransomware, zero-day exploits, and advanced persistent threats, has led to the widespread adoption of IDS in modern digital infrastructures. These systems analyze network traffic, detect anomalies or known attack signatures, and either alert security teams or take preventive actions. As cyberattacks become more frequent and damaging, computer security plays a vital role in safeguarding sensitive information and ensuring the uninterrupted operation of services. The increasing reliance on digital environments makes IDS an indispensable component of a cybersecurity framework.

As shown in Fig. 1 which illustrates the projected growth of the IDS market from 2025 to 2034, highlighting its increasing importance in the cybersecurity landscape [3]. The market size is expected to increase from USD 7.09 billion in 2025 to USD 20.18 billion in 2034, with a robust compound annual growth rate of 12.33%. As organizations continue to digitize their operations and store critical data online, IDS solutions play a crucial role in detecting and mitigating unauthorized access and malicious activity, making them essential for protecting sensitive information and maintaining operational integrity in an increasingly connected world.

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) play a pivotal role in defending against a significant portion of cybersecurity attacks. As cyber threats become more sophisticated and frequent, IDS and IPS technologies serve as frontline defenses by monitoring network traffic, identifying suspicious activities, and either alerting security teams or actively blocking malicious actions. They are

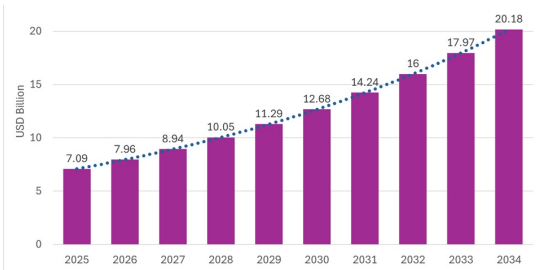

Fig. 1. The yearly growth of the Intrusion Detection and Prevention Systems Market

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especially critical in detecting complex threats such as zero-day exploits, advanced persistent threats, and insider attacks, which traditional security tools may fail to recognize. Given the growing reliance on digital infrastructure across industries, IDS and IPS form an essential component of a comprehensive cybersecurity strategy, helping organizations not only respond to known threats, but also adapt to emerging attack vectors in real-time. Their increasing deployment reflects their vital contribution to mitigating breaches, minimizing downtime, and protecting sensitive data from unauthorized access.

Detecting intrusions involves monitoring and analyzing system and network activity to identify signs of unauthorized access or malicious behavior. This process typically relies on Intrusion Detection Systems (IDS), which can be either signature-based or anomaly-based. Signature-based detection compares current activity to known patterns of malicious behavior, making it effective against previously identified threats. Anomaly-based detection, on the other hand, establishes a baseline of normal activity and flags deviations, helping to uncover novel or stealthy attacks. Modern approaches often incorporate machine learning and artificial intelligence to improve detection accuracy, reduce false positives, and adapt to evolving threats. Key techniques include monitoring traffic patterns, inspecting log files, examining user behavior, and correlating events from various sources. Effective intrusion detection also requires real-time alerts and integration with broader security frameworks to enable rapid incident response and minimize potential damage.

Conventional cybersecurity systems struggle with adaptability and generalization in dynamic environments. Additionally, the presence of data imbalance, high cardinality categorical features, and outliers in modern network datasets poses a significant challenge for both classical ML and DL models. The core problem addressed in this study is to design a hybrid, data-driven intrusion detection framework capable of achieving high accuracy while maintaining computational efficiency.

The primary objectives of this research are fivefold. First, to preprocess and transform the UNSW-NB15 dataset in accordance with best practices in data engineering, ensuring data quality and consistency. Second, to implement and compare a range of machine learning and deep learning models for binary classification of network traffic data. Third, to rigorously evaluate the performance of these models using both traditional train-test splits and k-fold cross-validation techniques. Fourth, to identify the most influential predictive features through the application of multiple feature selection methods. Finally, the study aims to recommend the most effective models for potential deployment in real-time cyber threat detection systems.

II. Related Work / Literature Review

Traditional machine learning methods have been extensively applied to intrusion detection problems [5]. Algorithms such as Logistic Regression, Decision Trees, Random Forests, and Naive Bayes are often favored for their interpretability and computational efficiency. Studies using the NSL-KDD, CIC-IDS2017 [17], and UNSW-NB15 datasets have shown that tree-based ensemble models like Random Forest and Gradient Boosting significantly outperform basic classifiers, especially in handling non-linear relationships and noisy data. However, many of these models struggle with class imbalance and high-dimensionality without appropriate preprocessing and feature selection.

Deep learning techniques such as Deep Neural Networks (DNN), Autoencoders, and Long Short-Term Memory (LSTM) networks have recently been used to improve detection rates in complex cyber attack scenarios. These models are capable of capturing high-level abstractions and complex temporal dependencies, which are often missed by shallow classifiers. Autoencoder-based anomaly detection has been utilized in unsupervised settings, particularly when labeled attack data is scarce. Despite their theoretical advantages, DL models can be sensitive to hyperparameter tuning and require substantial computational resources, which limits their practical deployment unless optimized.

Several comparative studies have benchmarked ML and DL models on public intrusion detection datasets. For instance, ensemble models like XGBoost and CatBoost have shown high classification performance and robustness. In contrast, basic DNNs without class balancing techniques often fail to generalize due to overfitting or bias towards the majority class. Few studies implement both traditional ML and advanced DL pipelines together, and even fewer compare them under consistent preprocessing and validation strategies. This study aims to bridge that gap by evaluating both paradigms within a unified framework and using consistent evaluation metrics and validation techniques such as 5-fold cross-validation. To better contextualize our work within the literature, Table I summarizes a selection of prior studies with the UNSW-NB15 dataset. It highlights the models used, performance metrics, and the key advantages and limitations of each approach.

A notable limitation in prior studies is the lack of integrated pipelines that combine comprehensive preprocessing (e.g., encoding, normalization, outlier handling), multiple feature selection techniques, and model explainability. Additionally, although many works evaluate individual ML or DL models, few attempt ensemble or hybrid modeling. Furthermore, the UNSW-NB15 dataset remains underutilized in deep learning contexts, particularly with methods like Focal Loss or Autoencoder-based anomaly detection. This study contributes to the literature by delivering a robust end-to-end system that tackles these gaps and compares a wide array of techniques across all modeling layers.

III. Materials and Methods

To provide a high-level overview of the methodology implemented, the overall pipeline from data ingestion to model evaluation is illustrated in Fig. 2. This diagram includes all stages applied throughout the study, from preprocessing and feature selection to model training and evaluation.

The overall methodological pipeline is visualized in Fig. 2.

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TABLE I
COMPARISON OF PREVIOUS STUDIES ON UNSW-NB15-BASED INTRUSION DETECTION

Year	Models Used	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Advantages	Disadvantages
2018 [6]	SVM, Naive Bayes, KNN, RF	85–87	86	84	85	Good classification , low overfitting	Slow on large datasets
2017 [7]	DNN, CNN, LSTM, Autoencoder	92–94	93	91	92	Ability to learn complex patterns	High computational cost, long training time
2016 [8]	CNN, GBM, RF, LR	90–95	91–94	89–92	90–93	Automatic feature extraction, fast learning	Requires preprocessing, hyperparameter tuning
2024 [9]	LSTM, BiLSTM, KNN, RF	91–93	92	90	91	Effective on time-series, memory capability	Complex structure, long training time
2018 [10]	LightGBM, Autoencoder + RF, XGBoost, SVM	90–94	91–93	89–91	90–92	Fast and efficient, effective anomaly detection	Complex models, parameter tuning required
2025 [11]	XGBoost, SVM, RF, LR, DT	94–96	95	93	94	High accuracy, low error rate	Model complexity, high computational cost
2019 [12]	LS-SVM, RF, LR, DT	93.3	98–100	-	-	Short training time, high precision	–

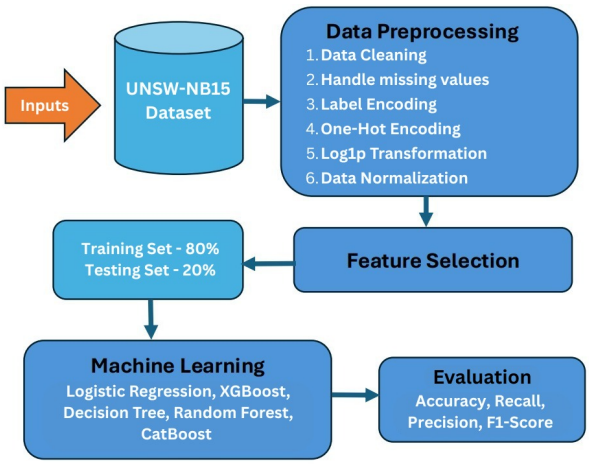

Fig. 2. Flow diagram of the machine learning pipeline used in this study.

A. Dataset Description

1) Data Source: The dataset used in this study is the publicly available **UNSW-NB15**, [13] developed by the Australian Centre for Cyber Security. It contains synthetic modern normal and attack behaviors. The dataset includes over **2.5 million records** with **49 features**, covering packet-level and flow-level characteristics.

2) Feature Description: The dataset offers a rich feature space composed of 49 attributes capturing various dimensions of network traffic. Table II provides a structured overview of these features, categorized for enhanced interpretability.

3) Features and Target Variable: The 49 features cover both quantitative and qualitative dimensions of network communication, such as packet size, flow duration, protocol type, service identification, and session state. This diverse representation facilitates comprehensive behavioral modeling of network activity.

The target variable, `Label`, is binary, where `0` denotes normal traffic and `1` indicates attack behavior. The organized structure of features—along with their distinct semantics—provided a strong foundation for designing a task-specific preprocessing pipeline, ensuring effective feature encoding and compatibility across both classical and deep learning models.

4) Data Preprocessing: To optimize model performance and reduce noise, features with high cardinality and minimal predictive value (e.g., IP addresses and port numbers) were excluded. Missing values were resolved using context-aware strategies such as row removal or statistical imputation.

Categorical variables were encoded based on their intrinsic meaning. Ordinal attributes like `state` underwent label encoding, while nominal features such as `proto`, `service`, and `ct_ftp_cmd` were one-hot encoded. To avoid sparsity, infrequent categories were grouped under an "Other" category. Skewed numerical features—such as `sbytes`, `dbytes`, `Sload`, and `Dload`—were transformed using $\log(x+1)$ to stabilize variance and normalize distributions. Subsequently, all numeric inputs were rescaled to the [0,1] interval using MinMax normalization to ensure uniform input ranges for machine learning models.

This structured and modular preprocessing pipeline not only enabled consistent input formatting but also enhanced generalizability and facilitated model interoperability, especially when benchmarking traditional ML models against deep learning architectures.

5) Categorical Feature Distribution: The dataset includes multiple categorical attributes such as protocol type, service category, and connection state. These attributes were analyzed to determine suitable preprocessing strategies based on their frequency distributions and class imbalance patterns.

The analysis revealed substantial imbalance among the categories, with a dominance of common protocol and service types such as TCP, UDP, DNS, and HTTP. Connection state values also showed strong concentration around a limited subset of outcomes.

To address this imbalance and improve model generaliza-

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TABLE II
COMPLETE FEATURE DESCRIPTIONS FROM THE UNSW-NB15 DATASET

Feature	Description	Feature	Description
srcip	Source IP address	res_bdy_len	Length of HTTP response body
sport	Source port number	Sjit	Jitter from source
dstip	Destination IP address	Djit	Jitter from destination
dsport	Destination port number	Stime	Session start time (timestamp)
proto	Protocol used (e.g., TCP, UDP, ICMP)	Ltime	Session end time (timestamp)
state	Final state of the connection	Sintpkt	Avg. time between source packets
dur	Duration of the flow (seconds)	Dintpkt	Avg. time between destination packets
sbytes	Bytes sent from source	tcprtt	TCP round-trip time
dbytes	Bytes sent from destination	synack	Time between SYN and ACK
sttl	TTL value from source	ackdat	Time between ACK and data
dttl	TTL value from destination	is_sm_ips_ports	Same source-destination IP/port
sloss	Packets lost at source	ct_state_ttl	Flows with same state and TTL
dloss	Packets lost at destination	ct_flw_http_mthd	Count of HTTP methods used
service	Detected service (e.g., HTTP)	is_ftp_login	Successful FTP login (1/0)
Sload	Source data rate (bps)	ct_ftp_cmd	Number of FTP commands
Dload	Destination data rate (bps)	ct_srv_src	Connections from source to service
Spkts	Packets sent from source	ct_srv_dst	Connections to destination/service
Dpkts	Packets sent from destination	ct_dst_ltm	Last 100 flows to destination IP
swin	Source TCP window size	ct_src_ltm	Last 100 flows from source IP
dwin	Destination TCP window size	ct_src_dport_ltm	From source IP to same dest port
stcpb	Source TCP base seq num	ct_dst_sport_ltm	To dest IP from same source port
dtcpb	Destination TCP base seq num	ct_dst_src_ltm	Same source-destination IP pair
smeansz	Avg. packet size from source	attack_cat	Attack category label
dmeansz	Avg. packet size from destination	Label	Class label (0 = Normal, 1 = Attack)
trans_depth	Depth of HTTP transactions		

tion, one-hot encoding was applied to nominal variables, while ordinal features exhibiting correlation with the target were processed using label encoding. Additionally, rare classes were grouped under a common "Other" label to mitigate sparsity and reduce the risk of overfitting.

B. Class Imbalance Handling

The dataset exhibits a class imbalance, with approximately 87% normal and 13% attack samples. To mitigate bias toward the majority class, class weighting was applied in the loss function during training. Additionally, stratified k-fold cross-validation ensured class proportions were maintained across all training and validation splits.

C. Feature Selection and Engineering

To detect multicollinearity and understand inter-feature relationships, Pearson correlation coefficients were computed among all numerical features. As visualized in Fig. 3,strong linear dependencies were observed between pairs like `sbytes` and `dbytes` (0.75), and `sttl` and `ct_state_ttl` (0.85), indicating redundancy and guiding selection decisions.

To quantitatively evaluate feature importance, a diverse set of ten selection methods was employed, including statistical tests (ANOVA F-test, chi-square), mutual information, recursive feature elimination (RFE), and embedded methods using Random Forest and XGBoost. Features such as `sttl`, `ct_state_ttl`, `sbytes`, and `dbytes` consistently ranked high across methods, highlighting their discriminative potential in intrusion detection.

Additionally, key preprocessing steps such as logarithmic transformation of skewed distributions, categorical encoding,

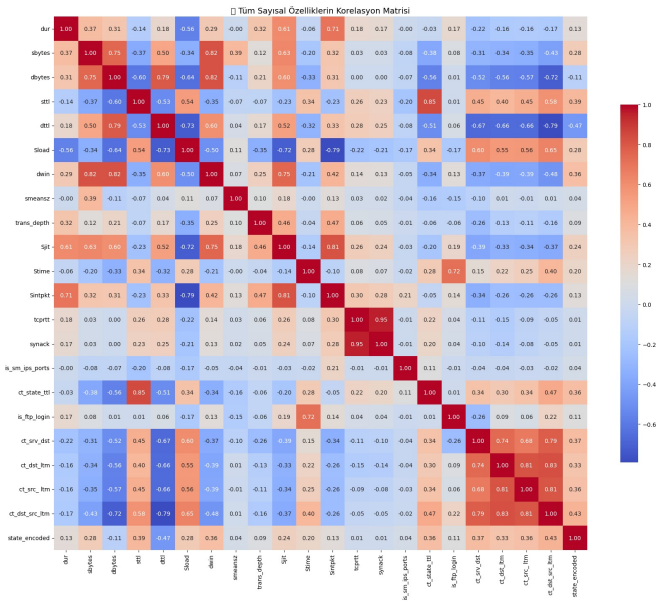

Fig. 3. Pearson correlation heatmap between numerical features and target variable.

and Min-Max normalization were treated as feature engineering strategies that enhanced learning efficiency and model robustness.

D. Machine Learning Algorithms

Five ML models were implemented:Logistic Regression, Decision Tree, Random Forest, XGBoost and CatBoost. Hyperparameters were optimized using `GridSearchCV` and algorithm-specific tuning methods. For example, `XGBoost`

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achieved effective performance with parameters such as `n_estimators=300`, `learning_rate=0.05`, and `max_depth=7`. These configurations ensured a balance between model complexity and generalization, helping reduce overfitting and improve prediction accuracy.

E. Model Evaluation Strategy

Both train/test split (80/20) and 5-Fold Cross-Validation were used to evaluate performance. Stratified splits ensured consistent class balance across folds. Key metrics included accuracy, precision, recall, F1-score, and confusion matrix. These metrics were computed during cross-validation to provide a robust assessment of model generalization. As illustrated in Fig. 4, the confusion matrix of the XGBoost model highlights its effectiveness in distinguishing between normal and attack traffic.

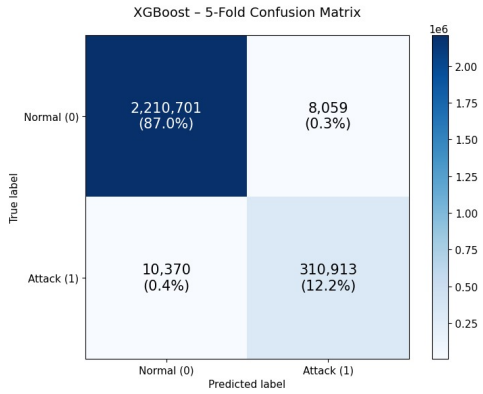

Fig. 4. Confusion matrix for XGBoost model (5-Fold CV).

IV. Experimental Results

Five machine learning and five deep learning models were trained and evaluated using accuracy, precision, recall and F1 score. The deep neural network based on focal loss [15] achieved the best overall performance, balancing F1 score and recall. Batch normalization and early stop configurations also performed competitively. Ensemble models like XGBoost and Random Forest maintained consistently high accuracy.

All ensemble models showed strong AUC values (>0.98), with Random Forest scoring the highest (0.9879). Logistic regression achieved the lowest AUC (0.8245), indicating limitations in high-dimensional scenarios. Table III summarizes the performance metrics of the five ML models under consistent conditions.

Artificial Neural Networks have proven effective in intrusion detection [14], with deeper DNNs offering improved accuracy and abstraction. Among DL models, the Focal Loss DNN achieved the highest minority-class recall. Batch-Normalized and Early-Stopping variants also showed high robustness. Table IV presents the evaluation of all DL models.

Tree-based models also performed well in minimizing false negatives, which is critical in real-time IDS. While the Autoencoder model—despite being unsupervised—achieved good recall, it may struggle with decision boundaries due to the lack of labeled data. Future work includes integrating explainable AI tools to analyze misclassified instances.

TABLE III
Performance Comparison of ML Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
LR	98.30	93.21	93.38	93.29
DT	99.40	97.83	97.43	97.58
RF	99.35	98.15	97.72	97.63
CatBoost	99.16	96.89	96.49	96.69
XGBoost	99.32	98.30	97.87	98.08

TABLE IV
Performance Comparison of DL Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
DNN (Baseline)	99.06	94.96	97.77	96.34
DNN + Early Stop	98.79	91.27	99.98	95.43
DNN + BatchNorm	98.91	92.51	99.47	95.86
DNN + Focal Loss	99.19	96.96	96.63	96.80
Autoencoder (AE)	94.73	72.92	92.84	81.68

V. Discussion

The comparative evaluation revealed that the Focal Loss-based DNN model achieved the best overall performance, especially in detecting samples from minority classes. Other deep learning configurations, such as batch-normalized and early-stopped DNNs, also showed competitive results. Among ML models, XGBoost delivered the highest F1-score, followed closely by Random Forest. The Autoencoder, despite being unsupervised, effectively captured anomaly patterns and achieved high recall, confirming its utility in scenarios with limited labeled data. This work utilized the modern and diverse UNSW-NB15 dataset. The integration of a complete preprocessing pipeline, feature engineering, and both supervised and unsupervised models under consistent evaluation settings offers a more comprehensive benchmark. The consistently top-ranked features (e.g., `sttl`, `ct_state_ttl`) can guide the design of lightweight monitoring tools. The successful use of class-weighted loss functions and Autoencoder-based detection highlights the potential of combining supervised and unsupervised methods for hybrid intrusion detection systems.

Detailed training and inference times for the evaluated machine learning models are presented in Table V. Furthermore, time efficiency analyses showed that models like CatBoost and XGBoost not only achieved strong predictive performance but also maintained short training and inference times, making them highly suitable for real-time deployment.

Similarly, deep learning models—particularly DNN variants with early stopping or batch normalization—demonstrated training times below 2 min and prediction speeds exceeding

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TABLE V
TRAINING AND PREDICTION TIME COMPARISON OF ML MODELS

Model	Training Time (s)	Total Pred. Time (ms)	Pred. Time (µs)
LR	32.3	43.4	50.0
DT	10.2	76.5	0.14
RF	276.8	3400.0	6.76
CatBoost	2.1	186.5	0.40
XGBoost	3.3	267.6	0.60

19,000 records per second. All experiments were carried out in a GPU-accelerated environment equipped with a T4 GPU, ensuring efficient parallel processing for training and inference tasks.

Table VI shows the timing performance of the deep learning models, including training durations and per-record inference speeds. These results indicate that with proper optimization, selected deep learning architectures can also be considered viable for scalable intrusion detection in time-sensitive environments.

TABLE VI
TRAINING AND PREDICTION TIME COMPARISON OF DL MODELS

Model	Training Time (s)	Pred. Time (µs/sample)
DNN (Baseline)	89.4	48.4
DNN + Early Stop	65.4	46.8
DNN + BatchNorm	48.6	50.9
DNN + Focal Loss	131.4	48.0
Autoencoder (AE)	106.2	49.5

This study was conducted with offline data; hence, real-time performance, latency, and concept drift were not explored. In addition, the models focused solely on binary classification. Future work should include multiclass setups and model explainability techniques like SHAP and LIME for improved interpretability. A limitation of this study is its focus on a single dataset. Although the results on UNSW-NB15 are promising, future work should test the proposed IDS framework on other benchmark datasets, such as CIC-IDS2017 [17] or NSL-KDD, to assess its generalizability in varying network conditions and attack types.

VI. CONCLUSION AND FUTURE WORK

This study developed a hybrid intrusion detection framework that leverages machine learning and deep learning models in the UNSW-NB15 data set. XGBoost and Focal Loss DNN achieved the best results with high accuracy and recall, effectively handling class imbalance and complex data patterns. Autoencoder-based detection also showed promise for unsupervised anomaly identification. Future directions include real-time deployment using stream-processing frameworks, multiclass classification of diverse attack types, explainability via SHAP or LIME, deep model enhancements (e.g., Transformers), and cross-dataset validation. These steps aim to build scalable, interpretable, and production-ready IDS solutions. The system has relatively long training and execution times. Therefore, since it deals with big data, the use of a GPU can be an effective solution to improve the efficiency of the proposed method, as also suggested in [16].

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